Coffee Sales Analysis

MySQL • CTEs • Window Functions • Business Insights

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Business Challenge

The coffee shop chain lacks a centralized dashboard or analytical report to monitor its performance. Management is unsure which hours are most profitable, which products are "dead stock" (slow-moving), and how different locations compare in terms of customer spending habits. They need to identify trends to prepare for seasonal shifts and marketing campaigns.

Key questions:

- Which products generate the most profit?
- When are peak hours for staffing decisions?
- What is the Month-over-Month (MoM) revenue growth percentage?
- What is the Average Transaction Value (ATV) across different store locations?

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Problem 1: Monthly Revenue Growth

What is the total revenue and total quantity sold for each month ?

SQL Query

```
SELECT
      date_format(transaction_date, '%b') as Month_Name,
   concat(Round(SUM(transaction_qty * unit_price),2)," $") as Total_Revenue,
   Sum(transaction_qty) as Total_Quantity
FROM Coffee
Group By Month_Name;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Month_Name	Total_Revenue	Total_Quantity
Jan	81677.74 $	24870
Feb	76145.19 $	23550
Mar	98834.68 $	30406
Apr	118941.08 $	36469
May	156727.76 $	48233
Jun	166485.88 $	50942

Insight

In the month of June,coffee shop is generating the maximum revenue with its highest of the sales.

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Problem 2: Location Benchmark

Calculate the total revenue, total transactions, and **Average Transaction Value (ATV)** for each store location. Which store is leading ?

SQL Query

```
SELECT
      store_location,
   CONCAT(ROUND(SUM(transaction_qty * unit_price), 2),' $') AS Total_Revenue,
   COUNT(DISTINCT transaction_id) AS Total_Transactions,
   CONCAT(ROUND(AVG(transaction_qty * unit_price), 2),' $') AS
Average_Transaction_Value
FROM
Coffee
GROUP BY Store_location
ORDER BY Total_Revenue DESC;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_location	Total_Revenue	Total_Transactions	Average_Transaction_Value
Hell's Kitchen	236511.17 $	50735	4.66 $
Astoria	232243.91 $	50599	4.59 $
Lower Manhattan	230057.25 $	47782	4.81 $

Insight

Hell's Kitchen Store is the leading coffee store among its other locations also Astoria store is very close to the leading sales.

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Problem 3: Revenue Contribution

What is the percentage contribution of each `product_category` **(Coffee, Tea, Bakery, etc.)** to the total revenue ?

SQL Query

```
With sales as
      (Select
            product_category,
            Concat(Round(sum(transaction_qty * unit_price),2)," $") as
Total_Revenue
   from coffee
   group by product_category
   order by Total_revenue)
Select
      product_category,
   Total_revenue,
   Round(sum(total_revenue) over (),2) as Overall_Sales,
   Concat(Round(Total_Revenue/sum(total_revenue) over () * 100, 2),' %') as
percentage_of_total
From Sales;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_category	Total_revenue	Overall_Sales	percentage_of_total
Loose Tea	11213.6 $	698812.33	1.6 %
Branded	13607 $	698812.33	1.95 %
Tea	196405.95 $	698812.33	28.11 %
Coffee	269952.45 $	698812.33	38.63 %
Coffee beans	40085.25 $	698812.33	5.74 %
Packaged Chocolate	4407.64 $	698812.33	0.63 %
Drinking Chocolate	72416 $	698812.33	10.36 %
Bakery	82315.64 $	698812.33	11.78 %
Flavours	8408.8 $	698812.33	1.2 %

Insight

The major of the sales is coming from coffee product category after it, tea is contributing the most.

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Problem 4: Sales Trends (MoM)

calculate the **Month-over-Month (MoM) growth** percentage in sales.

SQL Query

```
With sales as
      (select
      date_format(transaction_date," %b") as Month,
      Concat(Round(sum(transaction_qty * unit_price),2)," $") as Total_Revenue
from coffee
group by Month)
SELECT
      Month,
      total_Revenue as Current_Revenue,
      lag(total_Revenue) over () as Previous_month_Revenue,
      Concat(Round((total_Revenue - lag(total_Revenue) over () )/
lag(total_Revenue) over () * 100,2)," %") as Mom_growth_percentage
from sales;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Month	Current_Revenue	Previous_month_Revenue	Mom_growth_percentage
Jan	81677.74 $	NULL	NULL
Feb	76145.19 $	81677.74 $	-6.77 %
Mar	98834.68 $	76145.19 $	29.8 %
Apr	118941.08 $	98834.68 $	20.34 %
May	156727.76 $	118941.08 $	31.77 %
Jun	166485.88 $	156727.76 $	6.23 %

Insight

The coffee store has crossed more than 30 % growth rate in the month of may as its maximum month over month growth rate.

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Problem 5: The Morning Rush

How much revenue is generated during the peak morning hours (7:00 AM – 10:00 AM) compared to the rest of the day ?

SQL Query

```
SELECT
    CASE
        WHEN HOUR(transaction_time) BETWEEN 7 AND 9 THEN 'Morning Rush (7-
10 AM)'
        ELSE 'Other Hours'
    END AS time_period,
    SUM(transaction_qty * unit_price) AS total_revenue,
    COUNT(transaction_id) AS total_transactions,
    ROUND(SUM(transaction_qty * unit_price) / (SELECT
                SUM(transaction_qty * unit_price)
            FROM
                coffee) * 100,
            2) AS revenue_percentage
FROM
    coffee
GROUP BY time_period;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

time_period	total_revenue	total_transactions	revenue_percentage
Morning Rush (7-10 AM)	231395.86986500025	48846	33.11
Other Hours	467416.45974600315	100270	66.89

Insight

The store is generating more than 30 % of the revenue in the morning hours which indicates need of more staff on this hours.

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Problem 6: Peak Hour Analysis

Identify the top 3 busiest hours of the day for each store based on the number of transactions.

SQL Query

```sql
WITH HourStats AS (
    SELECT
        store_location,
        HOUR(transaction_time) AS hour_of_day,
        COUNT(transaction_id) AS total_transactions
    FROM coffee
    GROUP BY store_location, HOUR(transaction_time)
),
RankedHours AS (
    SELECT
        store_location,
        hour_of_day,
        total_transactions,
        RANK() OVER (PARTITION BY store_location ORDER BY total_transactions
DESC) as rnk
    FROM HourStats
)
-- Selecting only the top 3 hours per store
SELECT
    store_location,
    hour_of_day,
    total_transactions
FROM RankedHours
WHERE rnk <= 3
ORDER BY store_location, total_transactions DESC;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_location	hour_of_day	total_transactions
Astoria	10	5291
Astoria	9	5083
Astoria	8	4966
Hell's Kitchen	10	6957
Hell's Kitchen	8	6909
Hell's Kitchen	9	6767
Lower Manhattan	10	6297
Lower Manhattan	9	5914
Lower Manhattan	7	5792

Insight

8 to 10 A.M are the peak hours, where footfall is maximum as indicated by maximum no of transactions.

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Problem 7: Weekday vs. Weekend

Compare the average daily revenue generated on weekdays versus weekends. Should we adjust operating hours ?

SQL Query

```
SELECT
    CASE
        WHEN dayofweek(transaction_date) IN (1,7) Then "Weekend"
    Else "Weekdays"
    END AS Day_Type,
Round(sum(transaction_qty * unit_price)/count(distinct transaction_date),2) as
Average_Daily_Revenue
From coffee
Group by Day_Type;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Day_Type	Average_Daily_Revenue
Weekdays	3873.75
Weekend	3827.94

Insight

The store is generating a good ammount of more than 3500 $ as its average transaction value despite of its week type.

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Problem 8: Top 10 High-Performers

List the **Top 10** `product_detail` items by total revenue.

SQL Query

```
SELECT
   product_detail,
   ROUND(SUM(transaction_qty * unit_price), 2) AS Total_Revenue
FROM
   coffee
GROUP BY product_detail
ORDER BY Total_Revenue DESC
LIMIT 10;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_detail	Total_Revenue
Sustainably Grown Organic Lg	21151.75
Dark chocolate Lg	21006
Latte Rg	19112.25
Cappuccino Lg	17641.75
Morning Sunrise Chai Lg	17384
Latte	17257.5
Jamaican Coffee River Lg	16481.25
Sustainably Grown Organic Rg	16233.75
Cappuccino	15997.5
Brazilian Lg	15109.5

Insight

Sustainably Grown Organic Lg (large) is the leading product which is contributing to the maximum of sales.

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Problem 9: Menu Optimization

Identify the **bottom 10 products** by quantity sold. Should these be removed or promoted ?

SQL Query

```
SELECT
      product_detail,
   sum(Transaction_qty) as Quantity_Sold,
   Round(sum(transaction_qty * unit_price),2) as Total_Revenue
From coffee
group by product_detail
Order by quantity_sold
Limit 10;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_detail	Quantity_Sold	Total_Revenue
Dark chocolate	118	755.2
Spicy Eye Opener Chai	122	1335.9
Guatemalan Sustainably Grown	134	1340
Earl Grey	142	1270.9
Jamacian Coffee River	146	2883.5
Chili Mayan	148	1972.84
Columbian Medium Roast	148	2220
Primo Espresso Roast	150	3067.5
Lemon Grass	152	1360.4
Traditional Blend Chai	153	1369.35

Insight

Dark chocolate and spicy eye opener chai are the products which are having least of the sales which shows people don't like them so need to be removed.

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Problem 10: Category Deep Dive

Within the **'Coffee'** category, which product_type **(e.g., Drip coffee vs. Gourmet brewed coffee)** generates the most profit?

SQL Query

```
SELECT
   product_type,
   ROUND(SUM(transaction_qty * unit_price), 2) AS Total_Revenue
FROM
   coffee
WHERE
   product_category = 'Coffee'
GROUP BY product_type
ORDER BY Total_Revenue DESC
LIMIT 1;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_type	Total_Revenue
Barista Espresso	91406.2

Insight

Barista Espresso is the most demanded product type in the coffee category so its stock must be specially taken care of to meet the growing demand.

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Key Insights from Analysis

High-Value Product Dominance

The analysis of top-performing products shows that **Large-sized items** (specifically **Sustainably Grown Organic Lg** and **Dark Chocolate Lg**) are the primary revenue drivers.

Coffee Category Specialization

While the shop offers a diverse menu, **Barista Espresso** and **Gourmet Brewed Coffee** account for the vast majority of total coffee revenue. Premium and Organic brews have a loyal but smaller following.

The "Golden Hour" of Revenue

Across all store locations, the **10:00 AM hour** is the single most critical window for sales volume which shows there will be need of more staffing at this hour.

Footfall Consistency

Revenue data indicates an surprisingly **stable performance** between weekdays and weekends, with average daily revenue hovering around **$3,850** regardless of the day of the week.

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Recommendations

01

Dynamic Staffing Model

Implement a tiered staffing schedule. Increase barista count in **Lower Manhattan starting at 6:30 AM**, while keeping a leaner crew in **Astoria until 8:30 AM** to match the mid-morning peak.

02

Inventory Streamlining

Phase out "Small" cup sizes for top-selling coffees. The data shows customers prefer Regular and Large so, removing Small will reduce packaging costs and simplify workflow.

03

Peak Flow Management

Since **10:00 AM** is a universal peak, introduce a "Mobile Order Pickup" station to prevent counter congestion and maintain a high transaction speed during this window.

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Thank You

Gagandeep Singh

Tools : My SQL

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